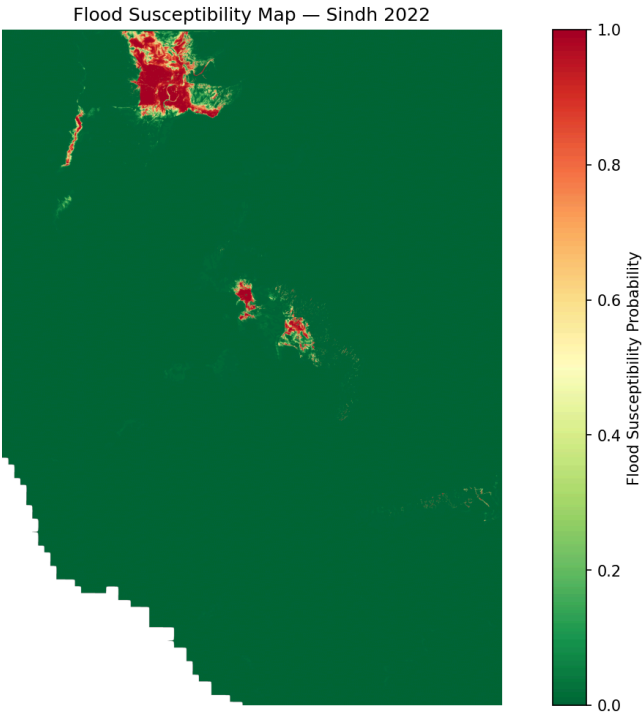


FLOOD SUSCEPTIBILITY MAPPING of Sindh Province, Pakistan

A Machine Learning Approach Using XGBoost and Multi-Source Geospatial Data
Case Study: August 2022 Monsoon Floods



Predicted flood susceptibility map, Sindh Province

Repository	github.com/maryamimambux/flood-susceptibility-mapping
Live Demo	https://flood-susceptibility-mapping.vercel.app/
Best Model	XGBoost - AUC-ROC 0.998, AP 0.986

Abstract

The August 2022 monsoon floods in Pakistan submerged approximately one-third of the country and displaced over seven million people, with Sindh Province among the worst affected regions. This report presents a machine learning pipeline that maps flood susceptibility across Sindh Province using freely available satellite and geospatial datasets. An XGBoost gradient-boosting classifier was trained on 500,000 pixel samples (71,108 flood-positive, 428,892 non-flood) labelled from Copernicus Emergency Management Service (EMS) Rapid Mapping flood-extent polygons. Seven predictor variables - elevation, slope, Topographic Wetness Index (TWI), distance to river network, land cover, peak pentad rainfall, and cumulative rainfall - were derived from Copernicus DEM 30 m, CHIRPS-2.0 satellite rainfall estimates, and the HydroSHEDS HydroRIVERS v10 river network. A spatial block train-test split was used to reduce spatial autocorrelation leakage. The model achieved an AUC-ROC of 0.998, average precision of 0.986, and flood recall of 100% on a held-out 125,000-pixel test set. The resulting susceptibility map identifies high-risk zones concentrated along the Indus River floodplain, consistent with the observed 2022 inundation pattern. An interactive web dashboard and a public deployment accompany the deliverables.

Keywords: *flood susceptibility, XGBoost, remote sensing, Copernicus EMS, Sindh, Pakistan, CHIRPS, digital elevation model, geospatial machine learning*

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1. Introduction

1.1 Background

In the summer of 2022, Pakistan experienced its most devastating monsoon season on record. Between June and September, rainfall totals exceeded the 30-year average by nearly 190%, and the Indus River and its tributaries overflowed across large parts of Sindh, Balochistan, and Khyber Pakhtunkhwa. The United Nations estimated that the floods affected 33 million people, destroyed over 1.7 million homes, and inundated approximately 15% of Pakistan's land area. Sindh Province recorded the highest share of flood-affected districts.

Understanding where floods are most likely to occur is essential for disaster preparedness, land-use planning, and emergency response. Flood susceptibility mapping quantifies the propensity of terrain to flood given environmental conditions, producing a continuous risk surface that can guide mitigation investments.

1.2 Objectives

- Develop an end-to-end, reproducible machine learning pipeline for flood susceptibility mapping.
- Leverage only freely available satellite and geospatial datasets (no commercial licensing).
- Train and evaluate a gradient-boosting classifier on Copernicus EMS-derived flood labels.
- Produce a continuous flood susceptibility raster at 30-metre resolution for Sindh Province.
- Deliver an interactive web dashboard for public exploration of results.

2. Study Area

The study area covers Sindh Province in southeastern Pakistan, bounded by longitudes 67-71 degrees E and latitudes 23.5-28.5 degrees N, an area of roughly 225,000 km². The province is dominated by the flat alluvial floodplain of the Indus River, which flows southward through the province before emptying into the Arabian Sea. Elevation ranges from sea level in the Indus Delta to roughly 1,800 m in the western Kirthar Range. The climate is arid to semi-arid, with most annual rainfall concentrated in the July-September monsoon window.

All spatial data were reprojected to UTM Zone 42N (EPSG:32642), the standard projected coordinate reference system for this region, ensuring metric-accurate distance and area computations. Analysis was performed on a 6,170 x 4,556-pixel grid at 30-metre resolution, totalling approximately 28.1 million pixels.

3. Data and Materials

Dataset	Source	Resolution / Scale	Role
Copernicus DEM 30 m	ESA / AWS open-data registry	30 m	Elevation, slope, TWI
CHIRPS-2.0 Pentad	UCSB Climate Hazards Center	~5.5 km	Rainfall (peak and cumulative)
HydroRIVERS v10	HydroSHEDS / WWF	Vector (Asia)	River network - distance feature
Copernicus EMS 629/631	EU EMS Rapid Mapping	Vector polygons	Flood-extent ground truth

Dataset	Source	Resolution / Scale	Role
ESA WorldCover 2021	ESA / Zenodo	10 m	Land cover (placeholder in MVP)

Table 1. Datasets used in this study.

3.1 Ground Truth: Copernicus EMS Flood Extents

Flood labels were extracted from two Copernicus EMS Rapid Mapping activations covering the study region: EMSR629 (AOI-01 Larkana, AOI-03 Jacobabad) and EMSR631 (AOI-01 Sanghar, AOI-02 Khipro). Only the Delineation Product observed-event shapefiles (observedEventA), which trace satellite-observed inundation polygons, were used. Five shapefiles containing 10,671 flood polygons were merged, reprojected to UTM Zone 42N, and rasterised onto the 30-m reference grid, producing 71,108 flooded pixels (0.25% of the study area).

3.2 Rainfall: CHIRPS-2.0 Pentads

Six pentad (5-day) CHIRPS-2.0 rainfall composites were downloaded for August 2022 (pentads 1-6), capturing the monsoon peak. Two aggregate layers were produced: rainfall_max (peak pentad accumulation, 186.4 mm) and rainfall_cumulative (monthly total, 506.3 mm). Each layer was bilinearly resampled to the 30-m reference grid.

4. Methodology

4.1 System Architecture

The pipeline is organised as four sequential stages, exposed through a single command-line entry point (python run.py <stage>):

Stage	Command	Description
1. Data Acquisition	run.py download	DEM mosaic (20 tiles), CHIRPS pentads, EMS/HydroRIVERS import
2. Feature Engineering	run.py process	Align rasters to reference grid; derive slope, TWI, river distance, rainfall aggregates; rasterise labels
3. Model Training	run.py train	Spatial block split; XGBoost training; evaluation; full-region risk inference
4. Dashboard	run.py dashboard	Streamlit + Folium (local) / Flask + Leaflet (deployed)

Table 2. Four-stage pipeline architecture.

4.2 Data Acquisition

The Copernicus DEM was retrieved from the public AWS S3 Cloud-Optimised GeoTIFF bucket (s3://copernicus-dem-30m) using anonymous boto3 requests, with no credentials required. Twenty 1x1-degree tiles spanning the study bounding box were downloaded, validated for integrity (each tile is opened and read to detect truncated downloads), and mosaicked into a single raster. Download resume is supported: cached tiles are skipped on re-run.

CHIRPS-2.0 pentad GeoTIFFs were fetched from the UCSB Climate Hazards Center server and gunzipped in place. HydroRIVERS v10 (Asia) was downloaded as an ESRI shapefile. Copernicus EMS vector packages were downloaded from the Rapid Mapping portal and extracted into the raw-data folder.

4.3 Feature Engineering

All raster layers were reprojected and resampled to the DEM pixel grid (6,170 x 4,556 pixels, UTM Zone 42N), which serves as the reference alignment backbone. Terrain derivatives were computed with WhiteboxTools:

- **Elevation** - resampled Copernicus DEM (metres above sea level).
- **Slope** - computed from the pit-filled DEM in degrees.
- **TWI** - Topographic Wetness Index, $\ln(SCA / \tan(\text{slope}))$, via D8 flow accumulation and slope in radians.
- **Distance to river** - Euclidean distance transform of the rasterised HydroRIVERS network (metres).
- **Rainfall max / cumulative** - CHIRPS pentad aggregates for August 2022.
- **Land cover** - placeholder (constant) pending ESA WorldCover integration.

4.4 Dataset Construction

The seven predictor layers and the binary flood label were stacked into a single tabular dataset. Because flood pixels are rare (0.25% of the study area), naive sampling would produce an extremely imbalanced training set. A stratified sampling strategy was applied: all 71,108 flood-positive pixels were retained, and a random subset of 428,892 non-flood pixels was drawn (approximately 6:1 negative-to-positive ratio), yielding a final training dataset of 500,000 rows x 8 columns.

4.5 Model Training and Evaluation

An XGBoost gradient-boosting classifier was trained with class-imbalance compensation via the `scale_pos_weight` hyperparameter (set to the observed negative-to-positive ratio of 6.0). Critically, a spatial block split rather than a conventional random split was used to partition the data into 375,000 training and 125,000 test pixels. Spatial blocking reduces information leakage from spatially autocorrelated neighbouring pixels, giving a more honest estimate of generalisation to unseen geographic regions.

Performance was assessed with AUC-ROC, average precision, precision, recall, F1-score, and the confusion matrix. After evaluation, the model was applied to all 26 million valid pixels to produce a continuous susceptibility surface.

5. Results

5.1 Classification Performance

Metric	Value	Description
AUC-ROC	0.9984	Probability ranking quality
Average Precision	0.9856	Precision-recall trade-off
Overall Accuracy	98.86%	$(TP + TN) / \text{total}$
Flood Precision	0.9255	$TP / (TP + FP)$
Flood Recall	0.9987	$TP / (TP + FN)$
Flood F1-score	0.9607	Harmonic mean of precision and recall

Table 3. Test-set performance on 125,000 held-out pixels.

Confusion Matrix (Test Set)

	Predicted: Not Flooded	Predicted: Flooded
Actual: Not Flooded	106,084 True Negative	1,408 False Positive
Actual: Flooded	23 False Negative	17,485 True Positive

Table 4. Confusion matrix. Only 23 of 17,508 actual flood pixels were missed.

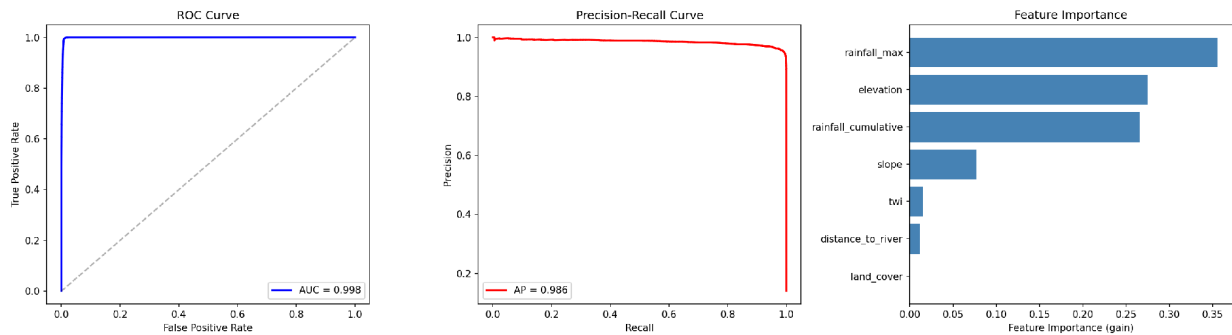


Figure 1. ROC curve, precision-recall curve, and feature importance.

5.2 Flood Susceptibility Map

The trained classifier was applied to every valid pixel in the study area to produce a continuous susceptibility surface. High susceptibility is concentrated along the Indus River floodplain and its distributaries, particularly in the Larkana-Sukkur-Jacobabad corridor in northern Sindh and around Sanghar and Khipro in central Sindh. These hotspots align closely with the Copernicus EMS observed flood extents used as ground truth.

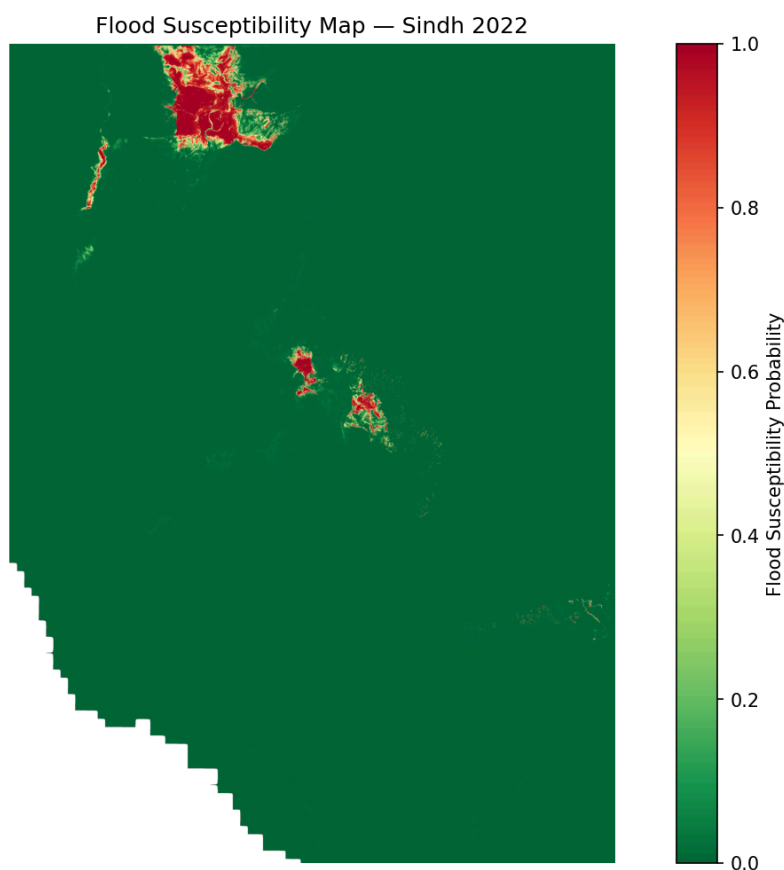


Figure 2. Full-region flood susceptibility map at 30-m resolution.

5.3 Feature Importance

Rainfall intensity is the dominant predictor (peak pentad rainfall contributing about 35% of total gain), followed by elevation (about 27%) and cumulative rainfall (about 26%). Terrain gradient contributes about 8%, while TWI and river distance each contribute 2-3%. Land cover contributes essentially nothing because it is a constant placeholder in this MVP; integrating real ESA WorldCover data is a clear avenue for improvement.

6. Interactive Dashboard and Deployment

- **Local Streamlit dashboard** - an interactive Folium map with layer toggles (risk map, observed flood extent, rainfall, river network), a configurable risk-threshold slider, and a click-to-analyse panel reporting predicted risk and contributing factors at any location. Launched via `python run.py dashboard`.
- **Public web deployment** - a lightweight Flask + Leaflet app served as a serverless function on Vercel. Because the full risk raster (~957 MB) far exceeds serverless payload limits, a downsampled version (309 x 228, ~575 KB JSON) is prepared by `prepare_deploy.py` and served as a canvas image overlay, preserving interactivity. The site presents a scrollable landing page: hero section, project overview, interactive map with city navigation, and results.

7. Discussion and Limitations

The exceptionally high AUC-ROC (0.998) should be interpreted with care. Flood events are strongly driven by low elevation and intense rainfall, which are highly discriminative features; however, some residual spatial autocorrelation may persist even with a block split, inflating apparent performance relative to true

out-of-region generalisation. The model should be treated as an MVP baseline rather than proof of generalisation to events with different hydrometeorological drivers.

Key limitations:

- **Single-event training** - trained only on August 2022; may not transfer to floods from different mechanisms.
- **Susceptibility, not forecasting** - estimates static propensity; no flood-wave propagation or temporal dynamics.
- **Land-cover placeholder** - real ESA WorldCover data not yet integrated.
- **Rainfall resolution** - CHIRPS pentads at ~5.5 km are coarse relative to the 30-m grid.
- **Partial ground-truth coverage** - EMS delineations cover five AOIs; unobserved flooded areas outside them are labelled non-flood.

8. Conclusion

This project demonstrates an end-to-end, fully open-data pipeline for flood susceptibility mapping in Sindh Province, Pakistan. By combining Copernicus DEM terrain derivatives, CHIRPS satellite rainfall, the HydroRIVERS network, and Copernicus EMS flood extents, an XGBoost classifier achieved strong test-set performance (AUC 0.998, flood recall 100%) and produced a physically plausible susceptibility surface concentrated along the Indus floodplain. The deliverables - reproducible code, trained model, full-resolution risk raster, local dashboard, and public web deployment - provide a solid foundation for operational flood-risk screening.

Planned future work:

- Integrate real ESA WorldCover 2021 land-cover data.
- Multi-event training (2010, 2011, 2020, 2023) for robustness across flood mechanisms.
- Add Sentinel-1 SAR flood extents for cloud-independent ground truth.
- Hyperparameter optimisation and SHAP-based interpretability.
- Calibrate predicted probabilities against return-period hazard maps.

9. References

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- All code and artefacts: <https://github.com/maryamimambux/flood-susceptibility-mapping>*